

South African Fraud Detection Using Machine Learning: Summary Report

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Project Type: Data Science / Machine Learning

Dataset: Synthetic (adapted to South African transaction behaviour)

1. Introduction

Fraudulent transactions present a significant challenge for financial institutions and digital commerce platforms in South Africa. As transaction volumes increase and payment systems become more digitised, traditional rule-based fraud detection systems struggle to adapt to evolving fraud patterns. Machine learning offers a flexible and data-driven approach to identifying suspicious activity, but its effectiveness is often constrained by data imbalance and regulatory requirements.

This project applies machine learning techniques to a synthetic dataset designed to reflect realistic South African transaction behaviour. The primary focus is not only on predictive modelling, but also on understanding the limitations of fraud detection under severe class imbalance and ensuring transparency through explainable model behaviour.

2. Business Understanding

The core business problem addressed in this project is the detection of rare fraudulent transactions within a large volume of legitimate activity. Fraud events are infrequent but carry high financial and reputational costs, making missed detections particularly undesirable. At the same time, excessive false positives can negatively impact customer experience and operational efficiency.

The objectives of the project are therefore to:

- Explore the challenges of fraud detection under extreme class imbalance
 - Evaluate trade-offs between fraud detection sensitivity and false alerts
 - Ensure model interpretability to support trust and regulatory compliance
 - Simulate a realistic South African fraud detection scenario
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3. Data Understanding

Due to privacy regulations such as POPIA and institutional confidentiality, real-world fraud datasets are rarely accessible. As a result, a synthetic dataset was used and adapted to reflect key characteristics of South African transaction behaviour, including transaction values, temporal patterns, and realistic fraud ratios.

The target variable, **is_fraud**, is binary:

- **1** indicates a fraudulent transaction
- **0** indicates a legitimate transaction

Initial data exploration revealed:

- Severe class imbalance, with fraud accounting for a small minority of transactions
 - A mix of numerical and categorical features
 - Transaction amount and time-based features with potential predictive relevance
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4. Data Wrangling and Exploratory Data Analysis (EDA)

Data wrangling included validation of data types, verification of missing values, and consistency checks across features. Exploratory data analysis was conducted to understand transaction behaviour and identify patterns relevant to fraud detection.

Key EDA insights included:

- **Class imbalance:** Fraud cases are heavily underrepresented, necessitating specialised handling
- **Transaction amounts:** Fraudulent transactions tend to cluster within specific value ranges
- **Temporal patterns:** Fraud occurrences show variation across time-based features
- **Feature relationships:** Low multicollinearity among numerical features supports model stability

EDA confirmed the presence of interpretable patterns, while also highlighting the inherent difficulty of minority-class detection.

5. Preprocessing and Feature Engineering

Categorical features were encoded to ensure compatibility with machine learning algorithms. The dataset was split into training and testing sets using stratified sampling to preserve class proportions.

To address class imbalance during training, **SMOTENC** was applied. This resampling technique is suitable for datasets containing both numerical and categorical variables and was used exclusively on the training data to avoid information leakage. Numerical features were scaled to support stable model optimisation.

6. Model Training

An **XGBoost** classifier was selected due to its effectiveness on tabular data and its ability to capture non-linear feature interactions. Hyperparameter tuning was performed using **GridSearchCV** with cross-validation to improve generalisability and reduce overfitting.

Key parameters tuned included learning rate, tree depth, and the number of estimators. This systematic approach ensured a well-regularised model rather than maximising performance at the expense of robustness.

7. Model Evaluation

Model performance was evaluated using metrics appropriate for imbalanced classification problems:

- Precision
- Recall
- F1-score
- Confusion matrix
- ROC-AUC
- Precision–Recall AUC

The results show that high overall accuracy is driven primarily by correct classification of non-fraudulent transactions. Fraud recall remains low, even after probability threshold tuning, highlighting the difficulty of detecting rare events with limited minority-class data. Threshold adjustment primarily increases false positives without materially improving fraud detection sensitivity.

The ROC-AUC indicates weak discriminative capability, consistent with overlapping class distributions. In contrast, Precision–Recall AUC provides a more informative evaluation of minority-class performance and better reflects the model’s practical limitations.

8. Explainability with SHAP

To enhance transparency, SHAP (Shapley Additive Explanations) was used to interpret model predictions. SHAP analysis identified transaction amount and time-based features, such as hour-of-day and weekend indicators, as influential

contributors to model output. These findings are consistent with insights observed during exploratory data analysis and align with domain expectations.

While overall predictive performance is limited, SHAP provides valuable local, transaction-level explanations that support model interpretability, auditability, and responsible use in regulated environments.

9. Conclusion

This project demonstrates the practical application of machine learning to fraud detection within a transaction environment inspired by the South African financial ecosystem. The findings highlight both the potential and the limitations of data-driven fraud detection under severe class imbalance.

Although resampling and threshold tuning were applied, minority-class performance remains constrained by data scarcity rather than model complexity. The use of explainability techniques supports transparency and responsible deployment, positioning the model as a decision-support tool rather than a fully automated fraud detection system. Overall, the project provides a realistic and methodologically sound foundation for further research and improvement.

10. Recommendations

- Deploy the model as a **decision-support system** with human oversight
- Prioritise the collection of richer and more representative fraud data
- Incorporate behavioural and contextual features where legally permissible
- Combine machine learning outputs with rule-based fraud detection systems
- Explore cost-sensitive learning and business-driven evaluation framework